Yulin Shao

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Research Interests

Causal inference and clinical trials methodology; uncertainty quantification and Bayesian methods; machine learning with statistical foundations, including large language models. Broadly interested in developing rigorous statistical methods at the intersection of theory and applied problems in data-driven sciences.

Education

Purdue University West Lafayette, IN
Ph.D. in Statistics Sep. 2026 – Present

University of Michigan, Ann Arbor Ann Arbor, MI
M.S. in Biostatistics Sep. 2024 – May 2026

- GPA: 4.2/4.3 (Top 3%; A+ = 4.30)

University of Michigan, Ann Arbor Ann Arbor, MI
B.S. in Honors Mathematics & Data Science Sep. 2022 – May 2024

Macalester College *(Transferred to University of Michigan)* St. Paul, MN
Honors Mathematics & Computer Science Sep. 2020 – May 2022

- GPA: 4.0/4.0 (Top 1%)

Publications & Manuscripts

- **Shao, Yulin**, et al. "How should covariates be handled in randomized trials? Empirical evidence from 50 trials and recommendations for practice." *Journal of Clinical Epidemiology* (2026): 112374. `doi:10.1016/j.jclinepi.2026.112374` `arXiv:2602.00434`

Research Experience

University of Michigan, Ann Arbor Ann Arbor, MI
Graduate Research Assistant / *Supervisor: Prof. Bingkai Wang* May 2025 – May 2026
Empirical Study of Statistical Methods for Randomized Clinical Trials

- Curated and standardized 50 real-world RCTs from public repositories (29,094 participants; 574 treatment-outcome comparisons) with unified R pipelines for missing-data imputation, outcome filtering, and covariate standardization.
- Implemented and evaluated 18 statistical methods combining 6 modeling approaches (ANCOVA, ANHECOVA, IPW, logistic g-computation, TMLE, DML) with 3 covariate-selection strategies; showed ANCOVA achieves 17% average variance reduction while ML methods provide no empirical advantage in finite samples.
- Established evidence-based guidelines for covariate adjustment; published all datasets and reproducible code on GitHub to support clinical-trials methodology research.

University of Michigan, Ann Arbor Ann Arbor, MI
Graduate Research Assistant / *Supervisor: Dr. Wen Ye* Jan. 2025 – May 2025
Development of Diabetic Kidney Disease Sub-Model within the Michigan Diabetes Model

- Processed 160,000+ longitudinal observations from 1,948 DPPOS Phase 3 patients (up to 25+ years of follow-up) tracking 15+ clinical parameters (eGFR, UACR, A1C, etc.) in R.
- Built and validated a DKD progression sub-model using competing-risks analysis for CKD and

mortality and Kaplan-Meier estimators, enabling simulation of 7 major clinical outcomes for type-2 diabetes patients.

- Authored comprehensive data dictionaries for longitudinal DPP/DPPOS datasets, contributing to a cost-effectiveness framework for healthcare policy decisions.

University of Michigan, Ann Arbor Ann Arbor, MI
Undergraduate Research Assistant / *Supervisor: Prof. Colin Fogarty* Apr. 2023 – Dec. 2023
Development of a Prepivoted Permutation Test R Package

- Implemented Gaussian and bootstrap prepivoting approaches into a production-ready R package for hypothesis tests exact under equality of distributions and asymptotically valid for equality of parameters.
- Engineered high-performance parallel infrastructure via Rcpp to handle nested resampling (999 permutations $\times$ 200–500 bootstrap draws), achieving substantial speedup for Hotelling's T^2, ANOVA F-statistics, and maximum absolute t-statistics.
- Created extensive documentation and unit tests to ensure reproducibility and facilitate adoption.

Teaching Experience

Purdue University West Lafayette, IN
Graduate Teaching Assistant, Department of Statistics Aug. 2026 – Present

University of Michigan, Ann Arbor Ann Arbor, MI
Graduate Student Instructor (GSI), Department of Biostatistics Fall 2025

- BIOSTAT 600: Introduction to Biostatistics; BIOSTAT 625: Computing with Big Data

University of Michigan, Ann Arbor Ann Arbor, MI
Teaching Assistant, Department of Mathematics Fall 2023

- MATH 451: Advanced Calculus

Macalester College St. Paul, MN
Teaching Assistant, Department of MSCS Fall 2021, Winter 2022

- COMP 123: Core Concepts in Computer Science

Service

Curriculum Committee Member Fall 2025 – Winter 2026
Department of Biostatistics, University of Michigan

Awards & Honors

University Honors, University of Michigan 2023
Finalist Award (Top 1%), Mathematical Contest in Modeling 2022 2022
Turck Honors Scholarship, Macalester College 2020–2022
Dean's List, Macalester College 2020–2022